Physics Revision Notes

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Chapter 1

Particles and Radiation

1.1 The Atom

An atom is made up of a nucleus (a core of protons and neutrons) surrounded by electrons shells. This nucleus makes up most of the mass of the atom, but is very small compared to the electrons shells around it. Electrons are in electrons shells around the nucleus, representing different energy values of the electrons.

Protons are positively charged and electron are negatively charged, therefore, for atoms to be neutral, there must be an equal number of protons and electron. The mass of an atom is given (in relative atomic mass units) by the number of protons plus the number of neutrons, since the mass of electron are negligable at this scale. An atom of element X can be written as its relative atomic mass (written A or A_r) and atomic number (written Z) (the number of protons). From its mass and atomic number, the number of neutrons can be found by subtracting the number of protons from the total mass. The number of electrons must be the same as the number of protons, since atoms are neutrally charged. The notation for the atom in this form is:

$$^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \tag{1.1}$$

For example, uranium-235 (a uranium atom (element number 92, symbol U) with a relative atomic mass of 235) would be written as:

$$^{235}_{92}\mathrm{U} \tag{1.2}$$

1.1.1 The Nucleus and Strong Nuclear Force

1.2 Radioactive Decay

1.2.1 Alpha Decay

Alpha decay is where an unstable nucleus has to lose mass to become stable, so it releases an alpha particle (${}^{2}_{4}\alpha$), which is a particle made of 2 nucleon and 2 neutron, equivalent to a helium-2 nucleus (${}^{4}_{2}\mathrm{He}^{2+}$). This reduces the mass of the original atom by 4, and it loses 2 protons, changing its element. It will also leave the atom as an ion, which is called ionisation. Since the atom is losing 2 protons, the atom will have its atomic number decreased by 2.

This is the equation for alpha decay of element X to element Y:

$$ {}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \longrightarrow {}^{\mathrm{A}-2}_{\mathrm{Z}-4}\mathrm{Y} + {}^{4}_{2}\alpha \tag{1.3} $$

This can also be written with the alpha particle explicitly written as a helium nucleus:

$$ {}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \longrightarrow {}^{\mathrm{A}-2}_{\mathrm{Z}-4}\mathrm{Y} + {}^{4}_{2}\mathrm{He} \tag{1.4} $$

An example of this is the alpha decay of uranium-238 to thorium-234:

$$ {}^{235}_{92}\mathrm{U} \longrightarrow {}^{231}_{90}\mathrm{Th} + {}^{4}_{2}\alpha \tag{1.5} $$

1.2.2 Beta Minus Decay

Beta minus decay is where a neutron turns into a nucleon, releasing an electron (also called a beta minus particle) and an electron antineutrino. Inside the decaying neutron, a down quark becomes an up quark, which releases the electron to conserve charge and the electron antineutrino to conserve lepton/electron number (L_{e}). Since the atom gains a nucleon (and thus +1 charge), it becomes an ion, which is called ionisation. However, whereas alpha decay decreases the atomic number by 2, beta minus decay increases it by 1.

The equation for beta minus decay of element X to element Y is:

$$ {}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \longrightarrow {}^{\mathrm{A}+1}_{\mathrm{Z}}\mathrm{Y} + {}^{0}_{-1}\beta^{-} + \bar{\nu}_{\mathrm{e}} \tag{1.6} $$

This can also be written with the beta particle explicity written as an electron:

$$ {}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \longrightarrow {}^{\mathrm{A}}_{\mathrm{Z}+1}\mathrm{Y} + \mathrm{e}^{-} + \bar{\nu}_{\mathrm{e}} \tag{1.7} $$

An example of beta minus decay is the beta minus decay of hydrogen-3 to helium-3:

$$ {}^{3}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + {}^{0}_{-1}\beta^{-} + \bar{\nu}_{\mathrm{e}} \tag{1.8} $$

The force responsible for beta minus decay is the weak nuclear force, mediated by the W^{-} boson.

Since a neutron is turning into a nucleon, the equation for beta minus decay can be written using the quark flavour of the neutron and nucleon, as:

$$ \mathrm{udd} \longrightarrow \mathrm{uud} + \beta^{-} + \bar{\nu}_{\mathrm{e}} \tag{1.9} $$

Which can be further simplified by removing the u and d that are left unchanged, to just leave:

$$ \mathrm{d} \longrightarrow \mathrm{u} + \beta^{-} + \bar{\nu}_{\mathrm{e}} \tag{1.10} $$

1.2.3 Beta Plus Decay

Beta plus decay is very similar to beta minus decay, however instead of a neutron becoming a nucleon, a nucleon becomes a neutron, releasing a positron (also called a beta plus particle), and an electron neutrino. Inside the decaying nucleon, an up quark becomes a down quark, which releases the positron to conserve the charge and the electron neutrino to conserve lepton/electron number (L_e). Since the atom loses a nucleon (and thus -1 charge), it becomes an ion, which is called ionisation. However, whereas alpha decay decreases the atomic number by 2, and beta minus decay increases it by 1, beta plus decay decreases it by 1. The equation for beta plus decay of element X to element Y is:

$$ {}^{A}_{Z}\mathrm{X} \longrightarrow {}^{A-1}_{Z}\mathrm{Y} + {}^{0}_{1}\beta^{+} + \nu_e \tag{1.11} $$

This can also be written with the beta particle explicity written as an positron:

$$ {}^{A}_{Z}\mathrm{X} \longrightarrow {}^{A}_{Z-1}\mathrm{Y} + \mathrm{e}^{+} + \nu_e \tag{1.12} $$

An example of this is the beta plus decay of hydrogen-3 to helium-3:

$$ {}^{23}_{12}\mathrm{Mg} \longrightarrow {}^{23}_{11}\mathrm{He} + {}^{0}_{1}\beta^{+} + \nu_e \tag{1.13} $$

The force responsible for beta plus decay is the weak nuclear force, mediated by the W^{+} boson.

Since a nucleon is turning into a neutron, the equation for beta plus decay can be written using the quark flavour of the nucleon and neutron, as:

$$ \mathrm{uud} \longrightarrow \mathrm{udd} + \beta^{+} + \nu_e \tag{1.14} $$

Which can be further simplified by removing the u and d that are left unchanged, to just leave:

$$ \mathrm{u} \longrightarrow \mathrm{d} + \beta^{+} + \nu_e \tag{1.15} $$

1.3 Feynman Diagrams

A Feynman diagram represents an interaction between particles. It has a time axis, showing the progression of the interaction from the incoming particles to the outgoing particles.

This is a simple Feynman diagram, representing two electrons (e^-) repelling each other. The interaction is mediated by the electromagnetic force, carried by a virtual photon (γ):

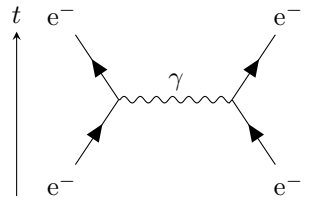


Figure 1.1: A simple Feynman diagram.

A virtual photon is a type of virtual particle. Virtual particles cannot be detected, but are put into models such as Feynman diagram to show force carriers such as photons and W boson. Any 'interal' particles (particles that don't enter/leave the diagram) are virtual particles in Feynman diagram.

Feynman diagram can be used to show more complicated processes, such as beta minus decay (note the virtual W^- boson):

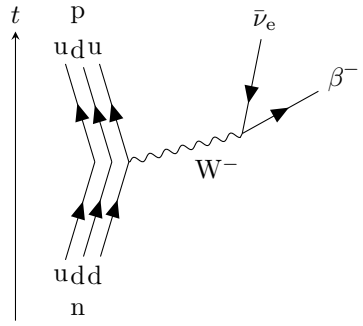


Figure 1.2: A Feynman diagram of beta minus decay; see section 1.2.2. Diagram source.

In Feynman diagram, fermions (leptons and baryons) are shown with solid straight lines with arrows representing their direction of travel through time. Matter travels forwards though time, and antimatter is modelled as travelling backwards, due to its charge being reversed from that of regular matter. Some

diagrams show antimatter as travelling forwards in time along with regular matter.

Most bosons (photons, W bosons, Z bosons) are shown with wavy lines (see the photons in figure 1.3 and the W^- boson in 1.3).

Mesons can be shown with a quark and antiquark line, or as a dashed line.

1.3.1 Examples

Beta Plus Decay

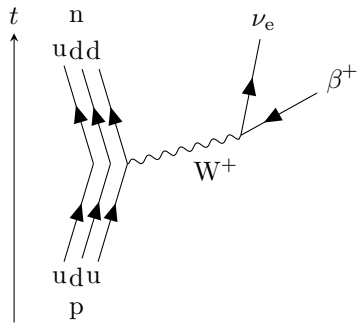


Figure 1.3: A Feynman diagram of β^+ decay; see section 1.2.3

Electron/Positron Annihilation and Pair Production

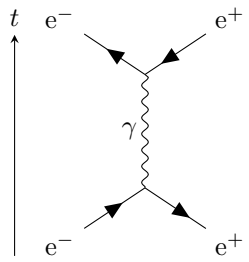


Figure 1.4: Electron/positron annihilation (bottom) and pair production (top).

1.4 Leptons

Leptons are fundamental particles that do not feel the strong nuclear force. The symbol for a lepton is ℓ. There are three generations of lepton, and each generation has a charged lepton and a neutrino variant.

The first generation has electrons (e^-) and electron neutrinos (ν_e).

The second generation has muons (μ^-) and muon neutrinos (ν_μ).

The third generation has muons (also called tauons; τ^-) and tau neutrinos (ν_τ).

The charged leptons all have a charge of -1, meaning that the charged antileptons all have charges of +1 (e^+, μ^+, τ^+). Antielectrons are called positrons. The neutrinos are uncharged, so antineutrinos ($\bar{\nu}$) are also uncharged.

1.4.1 Lepton Number

Lepton numbers is a quantum number describing whether or not a particle is a leptons. All regular matter leptons (not antileptons) have a lepton number of $L = +1$, and all antileptons have a lepton number of $L = -1$. All non-leptons have a lepton number of $L = 0$.

Each generation of lepton has its own lepton number variant; electrons and electron neutrinos (generation I lepton) have an electron number of $L_e = +1$; muons and muon neutrinos (generation II leptons) have a muon number of $L_\mu = +1$ and muons and tau neutrinos (generation III leptons) have a tau number of $L_\tau = +1$. All lepton numbers are conserved in interactions, meaning that something such as $\nu_e + n \longrightarrow p + e^-$ is valid ($L_e : 1 + 0 = 0 + 1$), but $\nu_e + n \longrightarrow \bar{p} + e^+$ cannot happen, due to electron number (L_e) not being conserved ($L_e : 1 + 0 \neq 0 - 1$).

1.5 Quarks

There are six types of quark, across three generations. The first generation contains up and down quarks (u, d), the second generation contains strange and charm quarks (s, c), and the third generation contains top and bottom quarks (t, b). Each generation gets progressively heavier. All quarks either have a charge of $Q = +\frac{1}{3}$ (up-type), or a charge of $Q = -\frac{2}{3}$ (down-type). All quarks have a baryon number of $B = +\frac{1}{3}$, and all quarks have a baryon number of $B = -\frac{1}{3}$.

1.5.1 Strange Quarks

1.6 Mesons

See section C.2.

Mesons are made up of a quark and an antiquark ($q\bar{q}$), and take the sum of the quantum numbers from the quarks, for example a meson containing a

strange quark and an up antiquark ($s\bar{u} \equiv K^-$) has a strangeness of $S = -1$ from the strange quark.

Pions

Pions are made of two generation I quarks (u and d). There are three types of pion:

- π^+ : $u\bar{d}$
- π^- : $d\bar{u}$
- π^0 : $\frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d})$

They represent the four different combinations two of up and down quarks, with the neutral pion (π^0) covering the up quark/up antiquark and down quark/down antiquark combinations. The neutral pion is in a superposition of $u\bar{u}$ and $d\bar{d}$, with each state being equally probable[1]. Due to the neutral pion always being a quark and its own antiquark, it almost immediately decays into two photons ($\pi^0 \longrightarrow 2\,\gamma$):

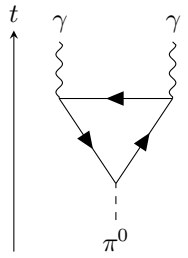


Figure 1.5: Neutral pion decay into 2 photons using a virtual quarks. Note that the photons are not virtual here, since they are emitted at the end of the interaction.

Kaons

Kaons are mesons made from a generation I quark and a strange quark. There are four types of kaon:

- K^+ : $u\bar{s}$
- K^- : $s\bar{u}$
- K^0 : $d\bar{s}$
- $\bar{K}^0$: $s\bar{d}$

[1] For some superposition $\psi = \alpha|x\rangle + \beta|y\rangle,\ a, b \in \mathbb{C}$, $P(x) = |\alpha|^2 \equiv \alpha\alpha^*$ and $P(y) = |\beta|^2 \equiv \beta\beta^*$, so for the neutral pion, $P(u\bar{u}) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$, which is the same as $P(d\bar{d}) = |-\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$, thus both states are equally probable.

1.7 Baryons

See section C.3.

A baryon is a particle made up of three quarks (qqq). Since all quarks have a baryon number of $B = +\frac{1}{3}$, a baryon has a baryon number of $B = \frac{1}{3}+\frac{1}{3}+\frac{1}{3} = 1$, since the quantum numbers from each quark are added. All baryons have unique antibaryons ($\overline{\text{qqq}}$), for example, an antiproton is $\overline{\text{uud}}$. All antibaryons have a baryon number of $B = -1$, since all antiquarks have a baryon number of $B = -\frac{1}{3}$. There are multiple different categories of baryons: N, Δ, Λ, Σ, Ξ, Ω.

1.7.1 Nucleons

Nucleons are a type of baryon made up of generation I quarks. There are two types of nucleon:

- Proton, p : uud
- Neutron, n : udd

Protons have a charge of $Q = +1$, and neutrons have a charge of $Q = 0$. Protons are the only stable baryon – every other baryon eventually decays into a proton; energy conservation laws and the Pauli exclusion principle stop neutrons in atoms from decaying[2].

1.7.2 Sigma Baryons

Sigma baryons (Σ) are another type of baryon. Sigma baryons contain two generation I quarks and a generation II or III quark. Ignoring charmed, top and bottom quark variants, there are three types of sigma baryon:

- Σ^+ : uus
- Σ^- : uds
- Σ^0 : dds

The variants for non-strange quarks are written as $\Sigma^{+/-/0}_{c/t/b}$, and have the same quark composition of the strange sigma baryons, but with the strange quark swapped for a charm quark/top quark/bottom quark.

[2] If a neutron in a nucleus were to decay to a proton, the proton would not be able to occupy low-energy states, since they are all taken by other protons and the Pauli exclusion principle forbids multiple protons in a nucleus from occupying the same energy state, so it would have to occupy a higher-energy state, which would require more energy than staying as a neutron, so conservation of energy forbids it. This is why neutrons only decay inside atomic nuclei (beta minus decay) if there are a lot more neutrons than protons, allowing there to be a lower-energy state for the neutron to decay into.

Chapter 2

Waves and Optics

Chapter 3

Mechanics and Materials

Chapter 4

Electricity

Chapter 5

Skills in Physics Year I

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Further Mechanics and Thermal Physics

Chapter 7

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Chapter 9

Turning Points in Physics

Chapter 10

Skills in Physics Year II

Appendix A

Appendix: Formula and Data Sheet

Appendix B

Appendix: Other Formulae

Appendix C

Appendix: Particle Zoo

C.1 Fundamental Particles

C.1.1 Quarks (q)

All quarks have non-zero mass and charge. They all have a baryon number of $B = +\frac{1}{3}$. All antiquarks ($\bar{\text{q}}$) have the opposite charge of their regular counterparts, and a baryon number of $B = -\frac{1}{3}$. The quarks are split into generations, u and d are generation I, s and c are in generation II and t and b are in generation III.

Quark	**Antiquark**	**Mass**/MeV	**Charge**/e	**Strangeness**
Up (u)	Antiup ($\bar{\text{u}}$)		$+\frac{2}{3}$	0
Down (d)	Antidown ($\bar{\text{d}}$)		$-\frac{1}{3}$	0
Strange (s)	Antistrange ($\bar{\text{s}}$)		$+\frac{2}{3}$	-1
Charm (c)	Anticharm ($\bar{\text{c}}$)		$-\frac{1}{3}$	0
Top (t)	Antitop ($\bar{\text{t}}$)		$+\frac{2}{3}$	0
Bottom (b)	Antibottom ($\bar{\text{b}}$)		$-\frac{1}{3}$	0

C.1.2 Leptons (ℓ)

All leptons have a non-zero mass. The charged leptons (e^-, μ^-, τ^-) all have a charge of -1; the neutrinos (ν_e, ν_μ, ν_τ) are all neutral. All leptons have a lepton number of $L = +1$, or more specifically, e^- and ν_e have an electron number of $L_\text{e} = +1$; μ^- and ν_μ have a muon number of $L_\mu = +1$ and τ^- and ν_τ have a tau number of $L_\tau = +1$. All antileptons ($\bar{\ell}$) have the opposite charge of their regular counterparts (this is obviously unchanged for neutrinos, since they are neutral), and their lepton numbers are $L_{\text{e}/\mu/\tau} = -1$.

Lepton	**Antilepton**	**Mass**/MeV	**Charge**/e
Electron (e^-)	Positron (e^+)		-1
Electron Neutrino (ν_e)	Electron Antineutrino ($\bar{\nu}_e$)		0
Muon (μ^-)	Antimuon (μ^+)		-1
Muon Neutrino (ν_μ)	Muon Antineutrino ($\bar{\nu}_\mu$)		0
Tau (τ^-)	Antitau (τ^+)		-1
Tau Neutrino (ν_τ)	Tau Antineutrino ($\bar{\nu}_\tau$)		0

C.1.3 Bosons

All bosons except the Higgs boson (H^0) mediate one of the fundamental forces:

- Photons (γ) mediate the electromagnetic force
- Gluons (g) mediate the strong nuclear force
- W and Z^0 bosons ($W^\pm$ and Z^0) mediate the weak nuclear force
- Gravitons (G) mediate gravity

Gravitons are hypothetical.

Boson	**Antiboson**	**Mass**/MeV	**Charge**/e
Photon (γ)	Photon (γ)		0
Gluon (g)	Gluon (g)		0
W Boson ($W^\pm$)	W Boson ($W^\mp$)		± 1
Z Boson (Z^0)	Z Boson (Z^0)		0
Higgs (H^0)	Higgs (H^0)		0
Graviton (G)	Graviton (G)		0

C.2 Mesons

There are 2 types of mesons required for A level physics: pions (π) and kaons (K). Kaons contain strange quarks (s) or antistrange quarks ($\bar{\text{s}}$), so they have a strangeness of ± 1.

Meson	**Quark Flavour**	**Antimeson**
Positive Pion (π^+)	$\text{u}\bar{\text{d}}$	Negative Pion (π^-)
Negative Pion (π^-)	$\text{d}\bar{\text{u}}$	Positive Pion (π^+)
Neutral Pion (π^0)	$\frac{1}{\sqrt{}}2(\text{u}\bar{\text{u}} - \text{d}\bar{\text{d}})$	Neutral Pion (π^0)
Positive Kaon (K^+)	$\text{u}\bar{\text{s}}$	Negative Kaon (K^-)
Negative Kaon (K^-)	$\text{s}\bar{\text{u}}$	Positive Kaon (K^+)
Neutral Kaon (K^0)	$\text{s}\bar{\text{d}}$	Neutral Antikaon ($\bar{\text{K}}^0$)
Neutral Antikaon ($\bar{\text{K}}^0$)	$\text{d}\bar{\text{s}}$	Neutral Kaon (K^0)

Meson	**Mass**/MeV	**Charge**/e	**Strangeness**
π^+		+1	0
π^-		-1	0
π^0		0	0
K^+		+1	-1
K^-		-1	+1
K^0		0	-1
$\bar{\text{K}}^0$		0	+1

C.3 Baryons

All baryons are made up of three quarks, meaning that they have a baryon number of $B = +1$. Antibaryons thus have a baryon number of $B = -1$, since they are made up of three antiquarks. Only nucleons (N) (C.3.1) and sigma baryons (Σ) (C.3.2) are shown below, but there are also delta, lambda, xi, and omega baryons (Δ, Λ, Ξ, Ω). There are possibly also 'exotic baryons' with more quarks and antiquarks; these are hypothetical and not also not shown below.

C.3.1 Nucleons (N)

Nucleons are the most common baryons, since they are the the components of atoms.

Baryon	**Quark Flavour**	**Antibaryon**
Proton (p/N^+)	uud	Antiproton ($\bar{p}/N^-$)
Neutron (n/N^0)	udd	Antineutron ($\bar{n}/N^0$)

Baryon	**Mass**/MeV	**Charge**/e	**Strangeness**
p		+1	0
n		0	0

C.3.2 Sigma (Σ)

Sigma baryons (Σ) are made up of two generation I quarks (u or d), and the third quark is either generation II or III (s, c, t, b). Only the strange sigmas are shown below.

Baryon	**Quark Flavour**	**Antibaryon**
Positive Sigma (Σ^+)	uus	Negative Antisigma ($\bar{\Sigma}^-$)
Negative Sigma (Σ^-)	uds	Positive Antisigma ($\bar{\Sigma}^+$)
Neutral Sigma (Σ^0)	dds	Neutral Antisigma ($\bar{\Sigma}^0$)

Baryon	**Mass**/MeV	**Charge**/e	**Strangeness**
Σ^+		+1	-1
Σ^-		-1	-1
Σ^0		0	-1

Glossary

alpha decay A type of radioactive decay where an unstable nucleus releases an alpha particle (2 protons and 2 neutrons, equivalent to a helium-2 nucleus). The equation for alpha decay is:

$$^{A}_{Z}X \longrightarrow ^{A-2}_{Z-4}Y + ^{4}_{2}\alpha \tag{C.1}$$

See section 1.2.1. 2, 3

alpha particle The particle released in alpha decay. It is made up of two protons and two neutrons, which is equivalent to a helium-2 nucleus. 2, 35

antilepton The antiparticle of a lepton. The charged antileptons are all positively charged. These are positrons (antielectrons, e^+), antimuons (μ^+) and antitau (τ^+). 6

antineutrino An uncharged antilepton with an extremely small mass and high velocity. Antineutrinos are the antiparticle of neutrinos. There are three generations of antineutrino, corresponding to the three generations of neutrino: electron antineutrino ($\bar{\nu}_e$, generation I), muon antineutrino ($\bar{\nu}_\mu$, generation II) and tau antineutrino ($\bar{\nu}_\tau$, generation III). Each generation has more mass than the previous. 2, 6, 37

antiquark The antiparticle of a quark. The charge is negated from its regular quark variant, meaning that up-type antiquarks have a charge of $Q = -\frac{2}{3}$, and down-type antiquarks have a charge of $Q = \frac{1}{3}$.. 5, 6, 36, 37

atom A nucleus with surrounding electron shells; makes up most matter. 1, 2, 3, 36, 37, 38

baryon A hadron made of three quarks, meaning that it has a baryon number of $B = +1$ (antibaryons have a baryon number of $B = -1$). 4, 35, 36, 37

baryon number A quantum number that determines whether a particle is a baryon. Baryons have a baryon number of $B = +1$ and antibaryons have a baryon number of $B = -1$. Quarks have a baryon number of $B = +\frac{1}{3}$ and antiquarks have a baryon number of $B = -\frac{1}{3}$, meaning that when

three quarks or quarks combine into a baryon/antibaryon, they give it a baryon number of $B = \pm 1$. 6, 35

beta minus decay A type of radioactive decay where a nucleus with too many neutrons has a neutron decay into a nucleon, releasing a beta minus particle (an electron) and an electron antineutrino. The neutron decaying into a nucleon can be simplified to just a single down quark in the neutron turning into an up quark in the nucleon. The equation for beta minus decay is:

$$\mathrm{d} \longrightarrow \mathrm{u} + \beta^{-} + \bar{\nu}_{\mathrm{e}} \tag{C.2}$$

See section 1.2.2. 2, 3, 4, 36

beta minus particle The name given to the electron released in beta minus decay. 2, 36

beta plus decay A type of radioactive decay where a nucleus with too many protons has a nucleon decay into a neutron, releasing a beta plus particle (a positron) and an electron neutrino. The neutron decaying into a nucleon can be simplified to just a single up quark in the nucleon turning into an down quark in the neutron. The equation for beta minus decay is:

$$\mathrm{d} \longrightarrow \mathrm{u} + \beta^{-} + \bar{\nu}_{\mathrm{e}} \tag{C.3}$$

See section 1.2.3. 3

beta plus particle The name given to the positron released in beta minus decay. 3, 36

boson A fundamental particle that obeys Bose-Einstein statistics. Bosons include gauge bosons such as photons, W bosons, Z bosons, and gluons as well as the Higgs boson (a scalar boson), gravitons (a hypothetical tensor boson) and mesons. All observed particles are either bosons or fermions. 5, 37, 38, 39

bottom quark The down-type generation III quark. 6, 38

charm quark The down-type generation II quark. 6, 38

down quark The down-type generation I quark. 3, 6, 38

electromagnetic force A fundamental force that acts on charged particles, repelling similar charges and attracting particles with opposite charges. 4, 38

electron The first generation charged lepton. It is the most common lepton, and is what balances the positive charge in atoms, forming electron shells. 1, 2, 4, 5, 6, 36, 37, 38

electron shell The different clouds of electrons around the nucleus of an atom. Each electron shell represents a different energy level of the electrons in them. An electron has the following quantum numbers: $L_e = 1$, $Q = -1e$; and a mass of approximately 9.109×10^{-31} kg, which gives a rest energy of 0.511×10^{-31} MeV, or 8.187×10^{-14} J. 1

fermion A fundamental particle that obeys Fermi-Dirac statistics. Fermions include quarks, leptons and baryons. All observed particles are either fermions or bosons. 4, 36, 38

Feynman diagram A diagram representing particle interactions. 4, 5

gluon A gauge boson that carries the strong nuclear force. 36

graviton A hypothetical tensor boson that mediates gravity. 36

hadron A particle made up of multiple quarks held together by the strong nuclear force. Hadrons include mesons and baryons. 35, 37

Higgs boson A scalar boson that allows other particles to have mass.. 36

ion An atom that has gained or lost an electron, meaning that the atom now has a non-zero charge. Positive ions are called cations, negative ions are called anions. 2, 3

lepton A particle that does not feel the strong nuclear force. There are two categories of lepton; the charged leptons and neutrino (uncharged leptons). There are three charged leptons (electrons, muons, muons), each with their own corresponding antilepton, and there are three neutrinos, each with their own corresponding antineutrino. 4, 6, 35, 37, 38

lepton number A quantum number that determines whether a particle is a lepton or not. Leptons have a lepton number of $+1$; antileptons have a lepton number of -1. Each generation of lepton has its own lepton number; electron number, muon number, tau number. Each generation of lepton number must be conserved in interaction. 3, 6

meson A hadron made up of a quark and an antiquark. 5, 36, 37

muon The generation II charged lepton. It has a negative charge, and its antiparticle (an antimuon, μ^+) has a positive charge. 6, 37

neutrino An uncharged lepton with an extremely small mass and high velocity. There are three generations of neutrino: electron neutrino (ν_e, generation I), muon neutrino (ν_μ, generation II) and tau neutrino (ν_τ, generation III). Each generation has more mass than the previous. 3, 6, 35, 37

neutron Makes up atomic nuclei along with protons. Neutrons are nucleons. See section C.3.1. 1, 2, 3, 35, 36, 38

nucleon The most common type of baryon, making up atomic nuclei. Protons and neutrons are the only nucleons. See section C.3.1. 38

nucleon Only stable type of baryon, making up atomic nuclei along with neutrons. Protons are nucleons. See section C.3.1. 1, 2, 3, 35, 36, 38

nucleus The core of an atom, made up of nucleons (protons and neutrons). 1, 2, 35, 36, 37, 38

pair production The process of a photon (with a minimum energy of 1.022MeV) producing an electron and a positron. See section 1.3. 5, 38

photon A gauge boson that carries the electromagnetic force. 4, 5, 36, 39

positron The antiparticle of an electron. It has a positive charge, and can annihilate an electron, or be produced alongside one in pair production. 3, 5, 6, 36, 38

quark Quarks are fermions that feel the strong force. They are fundamental particles. There are six types of quark, split over three generations: up (u, generation I), down (d, generation I), strange (s, generation II), charm (c, generation II), top (t, generation III) and bottom (b, generation III). The different quarks are also split into up-type and down-type, based on their charges (up-type has charge of $Q = \frac{2}{3}$; down-type has charge of $Q = -\frac{1}{3}$). The up-type quarks are: up, strange and top; the down-type quarks are down, charm and bottom. There is one up-type quark and one down-type quark per generation. 2, 3, 5, 6, 35, 36, 37

strange quark The up-type generation II quark. 6, 38

strong nuclear force The fundamental force that holds atomic nuclei together. At distances of $< \frac{1}{2}$ fm, it is an attractive force, overpowering the repulsive electromagnetic force and holding the protons in the nucleus together. At distances of 1 fm to around 3 fm it is repulsive, getting weaker and weaker, until it is almost 0 at around 3 fm. 6, 37

tau The generation III charged lepton. It has a negative charge, and its antiparticle (an antitau, τ^+) has a positive charge. 6, 37

top quark The up-type generation III quark. 6, 38

up quark The up-type generation I quark. 3, 6, 38

virtual particle A particle that helps mediate an interaction. The particle is not truly there, and cannot be observed, but is added to models to help explain interactions. 4, 39

virtual photon A photon that is a virtual particle. 4

W boson A gauge boson that carries the weak nuclear force. It is charged, either W^+ or W^-. The antiparticle of a W boson is a W boson with the opposite charge. 2, 3, 4, 5, 36

weak nuclear force The fundamental force that controls beta decay. 2, 3, 39

Z boson A gauge boson that carries the weak nuclear force. It is neutrally charged. Its antiparticle is itself. 5, 36